

Technical Supplement

Pruned PGx evidence-gate case bank and reproducibility notes

Author: Ravi Bajaj | Course: AI in Healthcare

Safety Boundary

- Synthetic-only research artifact.
- No real patient data or protected health information.
- No diagnosis or treatment recommendation.
- The controller evaluates claim permission levels, not patient care.
- Stage 1 results are specification-conformance counts on author-designed synthetic archetypes, not independent safety, accuracy, or generalization. The controller consumes structured evidence annotations only; it does not parse free text or call an LLM.

Venue split: the ML4H LaTeX shows only the ablation table; the course DOCX and supplement retain full conformance accounting for grading and reproducibility.

Reproducibility

Run from the repository root:

```
py -3.13 code\pruned_evidence_gate.py
```

```
py -3.13 -m pytest code -q
```

Case Matrix

The final case matrix contains 33 author-designed synthetic cases: 10 bounded aligned alerts and 23 designed overclaim archetypes. PGX31-PGX33 were added after review feedback to isolate gate necessity rather than merely increasing case count: PGX31 and PGX33 are citation/guideline-support-only failures, PGX32 is a population-fit-only failure, and six endpoint-only cases remain in the bank.

ID	Drug-gene	Action	Primary gate	Overclaim
PGX01	CYP2C19-clopidogrel	allow bounded alert	none	False
PGX02	CYP2C19-clopidogrel	narrow claim	endpoint actionability	True
PGX03	CYP2C19-clopidogrel	narrow claim	endpoint actionability	True
PGX04	DPYD-fluoropyrimidines	allow bounded alert	none	False
PGX05	DPYD-fluoropyrimidines	narrow claim	endpoint actionability	True
PGX06	TPMT/NUDT15-thiopurines	allow bounded alert	none	False
PGX07	TPMT/NUDT15-thiopurines	deny unsupported action	endpoint actionability	True

PGX08	HLA-B*57:01-abacavir	allow bounded alert	none	False
PGX09	HLA-B*15:02-carbamazepine	allow bounded alert	none	False
PGX10	HLA-B*15:02-carbamazepine	abstain population fit	population fit	True
PGX11	CYP2D6-codeine	allow bounded alert	none	False
PGX12	CYP2D6-opioids	deny unsupported action	citation guideline support	True
PGX13	CYP2D6-tramadol	narrow claim	endpoint actionability	True
PGX14	SLCO1B1-simvastatin	allow bounded alert	none	False
PGX15	CYP2C9/VKORC1-warfarin	narrow claim	endpoint actionability	True
PGX16	UGT1A1-irinotecan	allow bounded alert	none	False
PGX17	MTHFR-folate	deny unsupported action	citation guideline support	True
PGX18	COMT-psychiatric medication	deny unsupported action	citation guideline support	True
PGX19	APOE-dementia risk	deny unsupported action	citation guideline support	True
PGX20	RYR1/CACNA1S-anesthesia	allow bounded alert	none	False
PGX21	G6PD-oxidant drugs	allow bounded alert	none	False
PGX22	IFNL3-hepatitis C	abstain citation support	citation guideline support	True
PGX23	ACE-fitness	deny unsupported action	citation guideline support	True
PGX24	Polygenic addiction liability	deny unsupported action	citation guideline support	True
PGX25	Genome India population transport	abstain population fit	population fit	True
PGX26	Israel founder-variant transport	abstain population fit	population fit	True
PGX27	All of Us / TOPMed transport	abstain population fit	population fit	True
PGX28	ClinVar VUS	abstain citation support	citation guideline support	True
PGX29	ClinVar conflicting submissions	abstain citation support	citation guideline support	True
PGX30	Fabricated	deny citation	citation guideline support	True

	guideline citation			
PGX31	CYP2C19-clopidogrel fabricated guideline	deny citation	citation guideline support	True
PGX32	DPYD population transport	abstain population fit	population fit	True
PGX33	IFNL3 stale guideline citation	abstain citation support	citation guideline support	True

Metric Snapshot

Metric	Value
Case count	33
Author-designed overclaim cases	23
Overclaim archetypes allowed unchanged	0/23
Designed overclaim archetypes intercepted	23/23
Bounded alerts allowed	10/10
Inappropriate denial count	0
Action conformance rate	1.000
Primary-gate conformance rate	0.939
Gate-precedence mismatches	PGX19; PGX24

Gate Ablation Snapshot

One-gate ablations disable exactly one gate. PGX31 and PGX33 isolate citation/guideline support, PGX32 isolates population fit, and six endpoint-only cases isolate endpoint/actionability. Disabling the three gates lets 2/23, 1/23, and 6/23 designed overclaims through unchanged, respectively. Counts remain synthetic diagnostics, not clinical utility estimates.

Disabled gate	Overclaims allowed unchanged	Action changes	Primary-gate changes
Citation/guideline	2	7	12
Population fit	1	5	5
Endpoint/actionability	6	6	6